

Temporal Aggregation Erases the Advantage of Surgical-Domain Pretraining

Andrew Haggstrom

Abstract

Frozen vision foundation models are increasingly selected as perception backbones for surgical video systems, and surgical-domain pretraining is reported to confer large advantages over general-purpose encoders. We ask how much of that advantage survives the temporal aggregation a deployed system performs. Benchmarking four encoders spanning the pretraining 2×2 —general/medical $\times$ self-supervised/language-supervised—on Cholec80 phase recognition with paired video-level statistics over all 80 videos, we find that EndoViT leads DINOv2 by 6.0 accuracy points under frame-level linear probing ($p < 0.001$), but that 85% of this advantage disappears under causal temporal smoothing, leaving a residual gap statistically indistinguishable from zero ($+0.009$, $p = 0.29$). The effect is specific to the domain-pretrained encoder: every EndoViT gap shrinks by 55–85%, whereas the DINOv2–CLIP gap is unchanged ($+0.033 \rightarrow +0.032$, both $p < 0.001$), ruling out generic compression from smoothing. Our aggregator is a majority filter, substantially weaker than the multi-stage temporal convolutional networks used in practice, so this constitutes a lower bound on what a learned temporal head would recover. We additionally find that medical pretraining delivered by language supervision transfers essentially nothing (BiomedCLIP $\approx$ CLIP at every window, a well-powered null), that all four encoders over-segment surgical phase by 80–124 $\times$ at frame accuracies of 0.66–0.72, and that raw inter-frame drift is confounded by feature variance and must be normalized. We further report a result we retracted: an apparent rank inversion observed on a 32-video test split did not replicate at $n = 80$, which bears on how the sub-two-point encoder margins common in this literature should be read. Code, cached embeddings, and evaluation protocol are released.

1 Introduction

Frozen vision foundation models are now a default perception backbone for surgical video systems. The practical decision a group faces is which backbone to adopt, and increasingly whether to pay for surgical-domain pretraining (Batić et al., 2024; Jaspers et al., 2026; Park et al., 2026) over a strong general-purpose encoder such

as DINOv2 (Oquab et al., 2024).

That decision is made empirically, and the answer depends on the evaluation regime. Two coexist. The first is *frame-level probing*: linear probes and k -nearest-neighbour classifiers fit to frozen features from independently sampled frames, reported throughout the representation-learning literature as proxies for representation quality (Ramesh et al., 2023). The second is *frozen-backbone with temporal aggregation*: the two-stage protocol of TeCNO (Czempiel et al., 2020), in which cached frame-level features are passed to a multi-stage temporal convolutional network (Abu Farha and Gall, 2019). This is the protocol used for the frozen-backbone setting in recent surgical foundation model evaluations (Park et al., 2026; Spencer et al., 2026).

These are not the same measurement. The first is what most papers report when comparing encoders; the second is what a deployed system instantiates. The relationship between them—how much of a frame-level advantage survives aggregation, and whether that survival depends on how the encoder was pretrained—is, to our knowledge, unmeasured.

We measure it. Our contributions:

1. **The domain-pretraining advantage is largely temporal-recoverable.** EndoViT’s 6.0-point frame-level lead over DINOv2 ($p < 0.001$, 80 paired videos) shrinks by 85% under causal smoothing to a non-significant $+0.009$.
2. **The effect is specific to the domain-pretrained encoder.** All EndoViT gaps shrink 55–85%; the DINOv2–CLIP gap does not move. This localizes the effect to pretraining type rather than to smoothing itself.
3. **Supervision matters more than domain.** BiomedCLIP is indistinguishable from CLIP at every window; both self-supervised encoders beat both language-supervised ones.
4. **Magnitude.** All four encoders over-segment surgical phase by 80–124 $\times$ at frame accuracies that read as respectable.
5. **A lower bound.** A majority filter is far weaker than MS-TCN; that it already erases 85% bounds from below what a learned head recovers.

6. **A retracted result** and its implication for how small encoder margins in this literature should be interpreted (§5.2).

2 Related work

Frozen-feature evaluation and its fragility. Concerns about probe-based model comparison predate this application: Prior work has shown that models ranking poorly under linear probing can rank better under alternative readouts. Our observation that linear probing and k -NN select different encoders (§4.1) is a surgical instance of this known phenomenon and serves as motivation rather than as a novel finding. The reproducibility dimension has been documented sharply outside medicine: a recent audit of geospatial foundation models finds that papers reporting the same checkpoint on the same benchmark under the same nominal linear-probe protocol disagree by up to 12.7 points at the 90th percentile, largely because probe recipes go undisclosed.

Temporal models for surgical phase recognition. Since EndoNet (Twinanda et al., 2017) established the frame-level baseline, the field has advanced almost entirely through temporal modelling. TeCNO (Czempiel et al., 2020) introduced the two-stage protocol—frozen or fine-tuned frame features followed by a multi-stage temporal convolutional network (Abu Farha and Gall, 2019)—that remains the standard evaluation harness. Subsequent work replaced or augmented the aggregator: temporal memory relation networks (Jin et al., 2021), segment-level semantic modelling (Ding and Li, 2022), hybrid embedding-aggregation transformers (Trans-SVNet, Gao et al., 2021), long-video transformers (LoViT, Liu et al., 2025b), and online key-information transformers (SKiT, Liu et al., 2023). Reported accuracies now exceed 90%, against the 0.66–0.72 we measure for frozen frame-level features.

Framing. This progression is itself evidence for our premise: the architectures exist because frozen frame-level features are temporally unreliable. What has not been asked is whether the aggregation they supply *changes which backbone is best*. We show that it does, and specifically that it substitutes for domain pretraining. Funke et al. (2023) make a complementary argument at the metric level, showing that conclusions in this literature are sensitive to evaluation-metric choice; our results extend that sensitivity from the metric to the aggregation regime. These architectures exist *because* frozen frame-level features are temporally unreliable. We quantify what their aggregation recovers and show that this recovery substantially substitutes for domain-specific pretraining. Perez et al. (2026) report that a linear probe on frozen SurgCLIP features trails the specialised TAPIS architecture by only 2.34% F1 while out-

performing TAPIR and SlowFast, and that architectures jointly optimising backbone and temporal aggregation retain a 7.4% average advantage—which they characterise as modest relative to their substantially higher architectural complexity and training cost.

Surgical and medical foundation models. Surgical foundation models have proliferated rapidly and differ along two axes relevant here: pretraining corpus and supervision signal. Vision-only self-supervised models include EndoViT (Batić et al., 2024), which applies masked autoencoding (He et al., 2022) to Endo700k; EndoFM (Wang et al., 2023); GSViT (Schmidgall et al., 2024); and SurgeNetXL (Jaspers et al., 2026), trained on over 4.7M frames spanning four procedures. Vision–language models include PeskaVLP (Yuan et al., 2024) and SurgCLIP (Perez et al., 2026). ZEN (Park et al., 2026) unifies both families via multi-teacher distillation and reports the broadest evaluation to date—20 tasks under fine-tuned, frozen-backbone, few-shot and zero-shot settings. Outside surgery, BiomedCLIP (Zhang et al., 2025) provides medical-domain pretraining via language supervision on 15M biomedical figure–caption pairs, which we include as the medical/language-supervised cell of our 2×2 .

Reported domain-pretraining gains are protocol-dependent in a manner directly consistent with our finding. Jaspers et al. (2026) report procedure-specific pretraining improving Dice by 10.0–29.6% over ImageNet initialisation with *frozen* weights but only 2.8–12.8% with trainable weights—the frozen setting inflates the apparent benefit roughly two- to threefold on the fine-tuning axis, as we find it does on the temporal axis.

Most relevantly, Park et al. (2026) evaluate EndoViT on AutoLaparo (Wang et al., 2022)—a procedure absent from Endo700k—under the TeCNO protocol with an MS-TCN head, and report video-level F1 of 0.636 against 0.668 for a plain ImageNet-pretrained ViT-B. Outside its pretraining domain and under a stronger aggregator than ours, EndoViT’s advantage does not merely shrink but reverses. We return to this in §6. Reported gains are protocol-dependent in a manner consistent with our finding: Jaspers et al. (2026) report procedure-specific pretraining improving Dice by 10.0–29.6% over ImageNet initialisation with *frozen* weights but only 2.8–12.8% with trainable weights—the frozen setting inflates the apparent benefit roughly two- to threefold.

Closest prior work. Spencer et al. (2026) pair each of several visual encoders with the same MS-TCN++ temporal head under identical settings on SICS-155, comparing supervised encoders against self-supervised foundation models via a cached feature pipeline, and caution that deployment decisions should not rely on peak accuracy alone—noting that cataract-domain transfer yields mixed, backbone-dependent gains. That study

compares backbones *under* a fixed temporal head; we compare the same encoders *across* the frame-level and aggregated regimes and measure the change in each pairwise gap, with paired video-level intervals that distinguish a shrinking-but-real advantage from one that has become indistinguishable from zero.

3 Method

3.1 Encoders

We evaluate four encoders spanning the pretraining 2×2 (Table 1): DINOv2 ViT-S/14 (Oquab et al., 2024) (general, self-supervised), CLIP ViT-B/16 (Radford et al., 2021) (general, language-supervised), EndoViT ViT-B/16 (Batić et al., 2024) (surgical, self-supervised MAE), and BiomedCLIP ViT-B/16 (Zhang et al., 2025) (medical, language-supervised).

All encoders are frozen and used in evaluation mode. Each applies its own normalisation statistics—ImageNet, CLIP, endoscopic, and PMC respectively—as using incorrect statistics silently degrades features. Frames are resized to 224×224 with bicubic interpolation and anti-aliasing. We take the pre-projection pooled representation for both CLIP variants so that all four encoders are probed on comparable features. Implementations use `timm` (Wightman, 2019) and `open_clip` (Ilharco et al., 2021).

Embeddings are extracted once and cached as memory-mapped arrays partitioned by a hash of the encoder configuration, with SHA-256 manifests verified on every read; all downstream experiments are pure functions over this cache.

3.2 Evaluation protocols

P1, linear probe: class-balanced multinomial logistic regression on standardised frozen features. **P2, k -NN:** $k = 20$, cosine metric, no learned head. **P3, temporally smoothed accuracy:** a majority filter of window w applied to the P1 prediction stream. We report both *centered* (offline) and *causal/trailing* (real-time; the only deployable variant) filters; $w = 1$ recovers P1 exactly.

3.3 Temporal reliability metrics

All metrics are computed within a single contiguous sequence and never across video boundaries. *Normalized drift* is the mean inter-frame cosine distance divided by the sequence’s spread about its centroid; the normalisation is necessary rather than cosmetic (§5.1). *Boundary jitter* is the predicted phase-transition rate as a ratio to the ground-truth rate. *Conditional stability* multiplies stability by neighbour-label consistency, penalising the degenerate constant-output solution. *Velocity-flow coupling* is the Spearman correlation between embedding

Table 1: Encoders. All are ViTs at 224×224 . VRAM is peak allocation during extraction; SSL denotes self-supervised and “lang.” language supervision. Throughput is I/O-bound in our pipeline and is therefore not reported as model throughput.

Encoder	Pretraining	Dim	VRAM (MB)
DINOv2 S/14	LVD-142M, SSL	384	489
CLIP B/16	WIT-400M, lang.	768	1011
EndoViT B/16	Endo700k, MAE	768	856
BiomedCLIP B/16	PMC-15M, lang.	768	856

velocity and dense Farneback optical-flow magnitude, separating drift that tracks scene motion from drift that does not. Twelve synthetic controls with known answers gate the suite.

3.4 Data and statistics

We use Cholec80 (Twinanda et al., 2017): 80 laparoscopic cholecystectomy videos with per-frame phase annotations over seven phases and binary presence labels for seven instruments. Following the standard protocol we sample at 1 fps, yielding 184,498 frames. Phase labels are provided at 25 fps and instrument labels at 1 fps; we align them by taking the phase label at each sampled native frame index and assert index agreement rather than assuming it. Pilot experiments used CholecSeg8k (Hong et al., 2020).

Frames within a surgery are near-duplicates, so **all splitting and resampling is at the video level**. Our primary analysis is 5-fold cross-validation over all 80 videos with a shared fold assignment across encoders, so that every comparison is paired; each video is scored exactly once by a probe trained on the other four folds. Pairwise differences are assessed by a paired video-level bootstrap (Efron and Tibshirani, 1994) with 10,000 resamples of the per-video difference. Pairing cancels the large between-video variation in duration and difficulty; overlapping marginal intervals do not imply a non-significant paired difference. For comparability with published work we also report the fixed 40/8/32 split. Phase prevalence varies by up to $31\times$, so macro-F1 is reported alongside accuracy.

Our panel comprises $6 \text{ pairs} \times 5 \text{ windows} = 30$ tests, of which 24 are significant at $p < 0.05$ against ≈ 1.5 expected by chance; the null results below are therefore informative rather than merely underpowered.

4 Results

4.1 Frame-level protocols disagree

Table 2 shows two standard protocols selecting different winners. EndoViT has the best linear separability and

Table 2: Frame-level protocols, fixed 40/8/32 split. P1 selects EndoViT; P2 selects BiomedCLIP.

Encoder	P1 linear		P2 k -NN		gap
	acc	F1	acc	F1	
EndoViT	.704	.615	.522	.377	.238
DINOv2	.677	.597	.611	.463	.134
BiomedCLIP	.658	.578	.617	.504	.075
CLIP	.648	.568	.572	.391	.177

the worst geometric clustering: its phases are separable by a hyperplane but not by proximity, consistent with it having the smallest feature spread of the four (embedding standard deviation 0.425 versus 2.46 for DINOv2).

The per-phase pattern is identical across all four encoders: the long dissection phases score 0.69–0.77 F1 while transitional and rare phases collapse to 0.46–0.59. This failure is structural rather than model-specific, and concerns exactly the states a single frame cannot disambiguate.

4.2 Temporal reliability at $w = 1$

Every encoder over-segments dramatically. Predicted phase-transition rates exceed ground truth by $79.7\times$ (EndoViT), $110.5\times$ (DINOv2), $122.4\times$ (BiomedCLIP) and $124.3\times$ (CLIP), at frame accuracies of 0.66–0.72. The best encoder—a surgical foundation model pretrained on this data distribution—still announces eighty times too many phase transitions. This is a within-encoder measurement against ground truth, two orders of magnitude in size, and does not depend on any cross-encoder comparison.

4.3 The domain advantage converges away

Table 4 is the paper’s central result. EndoViT’s advantage over DINOv2 crosses from significant to non-significant between $w = 31$ ($p = 0.017$) and $w = 61$ ($p = 0.19$); at $w = 121$ EndoViT wins on 44 of 80 videos, a coin flip. Every EndoViT gap shrinks by 55–85%. No gap between general-purpose encoders shrinks at all: DINOv2–CLIP is $+0.033$ at $w = 1$ and $+0.032$ at $w = 121$, both $p < 0.001$.

This contrast is what makes the effect attributable to pretraining type. If smoothing merely compressed all differences toward a ceiling, the general-encoder gaps would shrink too. They do not. The interpretation we favour is that surgical pretraining principally buys single-frame discriminability, and that temporal aggregation recovers much of the same information from context—so the two routes to the same signal overlap.

Accuracy peaks near $w \approx 31$ under causal smoothing and declines thereafter, because a trailing window

necessarily lags every transition; macro-F1 peaks near $w \approx 15$ – 31 and falls below the unsmoothed baseline by $w = 121$ as short rare phases are erased. We therefore take $w \approx 31$ as the honest operating point. Under centered smoothing accuracy rises monotonically—an artifact of hindsight that the causal filter exposes.

Across the sweep, jitter falls 38 – $54\times$ while accuracy *rises*: the phase information was present in the frozen features throughout, and P1 gave no indication of how much temporal repair each encoder would require.

4.4 Supervision beats domain

BiomedCLIP and CLIP are statistically indistinguishable at every window $w \geq 15$ ($p = 0.17$ – 0.47 , differences ≤ 0.013 ; BiomedCLIP wins 38–47 of 80 videos). Medical-domain pretraining delivered through language supervision transfers essentially nothing to surgical video, whereas medical-domain self-supervised pretraining yields a large frame-level lead. Both self-supervised encoders beat both language-supervised encoders at every window. A plausible mechanism is that PMC-15M comprises static biomedical figures—histology, radiology, diagrams—whose visual statistics differ sharply from endoscopic video; nominal domain proximity does not imply transfer. Given that 24 of 30 comparisons in the same panel reach significance, this is a well-powered null.

4.5 Is drift signal or noise?

Spearman correlations between embedding velocity and optical-flow magnitude (10 test videos, all $p < 0.05$) are 0.499 (EndoViT), 0.471 (DINOv2), 0.451 (BiomedCLIP) and 0.388 (CLIP). Drift partly tracks scene motion, but roughly half the variance is unexplained by pixel motion, indicating a substantial spurious component.

Notably, this ordering is identical to the normalised-drift ordering and to the frame-level accuracy ordering. Three measurements that are methodologically independent—a label-free geometric statistic, a pixel-motion correlation, and a supervised probe—agree on the encoder ranking at $w = 1$. This makes the subsequent divergence under smoothing (§4.3) more striking: the disagreement is introduced by temporal aggregation, not present in the frozen features.

BiomedCLIP is the one encoder that separates from CLIP here (0.451 vs 0.388), despite indistinguishable phase accuracy (§4.4). Its embedding motion tracks scene motion better without that translating into semantic discriminability. A paired bootstrap over the same 10 videos confirms this difference is real ($+0.062$, 95% CI $[+0.043, +0.085]$, $p < 0.001$; BiomedCLIP higher on 9 of 10 videos). Four of the six pairwise flow comparisons are significant; the two that are not are DINOv2–EndoViT ($p = 0.069$) and DINOv2–BiomedCLIP ($p = 0.138$).

Table 3: Mean accuracy, 5-fold CV over all 80 videos, causal smoothing. The accuracy ordering is preserved at every window; the *gaps* are what change. Macro-F1 (below) reorders at $w \geq 61$, but not significantly (§4.3).

Encoder	$w=1$	15	31	61	121
EndoViT	.756	.805	.810	.795	.751
DINOv2	.697	.779	.790	.784	.742
BiomedCLIP	.676	.760	.770	.762	.723
CLIP	.664	.753	.764	.752	.710

Table 4: Paired accuracy differences, causal smoothing, $n = 80$ videos. Shrinkage is $1 - \Delta_{121}/\Delta_1$. All EndoViT gaps shrink; no general-encoder gap does. *** $p < .001$, ** $p < .01$, * $p < .05$, n.s. otherwise.

Pair	$w=1$	$w=31$	$w=121$	shrink
EndoViT–DINOv2	+.060***	+.019*	+.009	85%
EndoViT–BiomedCLIP	+.080***	+.040***	+.029**	64%
EndoViT–CLIP	+.092***	+.046***	+.041***	55%
DINOv2–CLIP	+.033***	+.026***	+.032***	1%
DINOv2–BiomedCLIP	+.021**	+.020*	+.020*	5%
BiomedCLIP–CLIP	+.012*	+.006	+.013	—

That DINOv2 and EndoViT do not separate on motion tracking is consistent with §4.3: the pair whose accuracy gap closes under aggregation is also the pair whose embedding dynamics are least distinguishable.

5 Analysis

5.1 Raw drift is variance-gameable

An early run indicated CLIP as *more* temporally stable than DINOv2 (raw inter-frame drift 0.042 versus 0.124). Diagnosis: CLIP’s feature spread was $3.5\times$ smaller and its embedding standard deviation $2.7\times$ smaller. Its drift was low because its features were low-variance, not because they were stable. The internal contradiction that flagged this was that CLIP simultaneously jittered *more* ($124\times$ versus $110\times$).

Normalising drift by sequence spread removes the artifact, and across four encoders the correction is not marginal but rank-changing. By raw drift the ordering is EndoViT (0.001), CLIP (0.043), BiomedCLIP (0.068), DINOv2 (0.124); normalised it becomes EndoViT (0.512), DINOv2 (0.665), BiomedCLIP (0.671), CLIP (0.800). **DINOv2 moves from worst to second-best.** An evaluation using raw inter-frame drift would rank the encoder that performs best under deployment smoothing as the least temporally stable of the four.

Two further observations support treating feature-space scale as a nuisance parameter. EndoViT has the *lowest* raw variance (emb_std 0.425) yet is genuinely the most stable once normalised—low variance realised well rather than badly. Conversely BiomedCLIP has the *highest* (3.202, above DINOv2’s 2.461) while CLIP has 0.913: a $3.5\times$ difference in feature-space scale between two encoders whose phase accuracy is statistically

indistinguishable (§4.4). Raw drift and raw variance each mislead in isolation; only the normalised metric distinguishes a bland encoder from a well-organised one.

5.2 A retracted result

On the fixed 40/8/32 split (32 test videos, 40 training), causal smoothing appeared to *invert* the DINOv2/EndoViT ordering: EndoViT +0.030 at $w = 1$, DINOv2 +0.021 at $w = 121$. We initially treated this as the paper’s principal finding.

It did not replicate. Under 5-fold cross-validation (80 test videos, 64 training) the sign never flips and EndoViT leads at every window. On the 32-video split, neither of the two comparisons we ran reached significance at any window, including EndoViT’s frame-level lead ($p = 0.076$). Both the evaluation-set size and the training-set size differ between the two analyses, so we do not attribute the discrepancy to either alone.

The implication extends beyond our own error: sub-two-point margins between frozen encoders on a 32-video Cholec80 test split are not resolvable, and such rankings are routinely reported in this literature without paired error bars. We recommend paired video-level bootstrapping and cross-validation over fixed splits whenever a reported margin is small.

6 Limitations

Domain overlap. Endo700k includes Cholec80, so EndoViT observed these videos during self-supervised pretraining. This inflates its frame-level lead and therefore makes the convergence result conservative. **Single dataset.** Cross-procedure replication, for which AutoLaparo (Wang et al., 2022) is the natural target, remains outstanding. **Aggregator strength.** A majority filter

is far weaker than MS-TCN (Abu Farha and Gall, 2019); our result is a lower bound, and a learned head would plausibly erase more of the domain advantage. **Panel size.** Four encoders suffice for pairwise claims at $n = 80$ videos but not for rank correlations. **Efficiency.** Reported throughput is pipeline-level and I/O-bound, not model-level.

7 Conclusion

Surgical-domain pretraining confers a large, unambiguous advantage on the frame-level probe the field uses to compare frozen backbones—6.0 accuracy points for EndoViT over DINOv2 across 80 paired videos. Under temporal aggregation, 85% of that advantage disappears and the remainder is indistinguishable from zero, while gaps between general-purpose encoders are untouched. Meanwhile every encoder tested over-segments surgical phase by 80–124 $\times$ at frame accuracies that read as respectable. Practitioners should evaluate frozen backbones under the aggregation regime they intend to deploy, report paired error bars on any margin below a few points, and treat frame-level gains from domain pretraining as an upper bound on deployed benefit.

References

- Y. Abu Farha and J. Gall. MS-TCN: Multi-stage temporal convolutional network for action segmentation. In *CVPR*, 2019.
- D. Batić, F. Holm, E. Özsoy, T. Czempel, and N. Navab. EndoViT: Pretraining vision transformers on a large collection of endoscopic images. *Int. J. Computer Assisted Radiology and Surgery*, 19(6):1085–1091, 2024.
- T. Czempel, M. Paschali, M. Keicher, W. Simson, H. Feussner, S. T. Kim, and N. Navab. TeCNO: Surgical phase recognition with multi-stage temporal convolutional networks. In *MICCAI*, pages 343–352, 2020.
- X. Ding and X. Li. Exploring segment-level semantics for online phase recognition from surgical videos. *IEEE Trans. Medical Imaging*, 41(11):3309–3319, 2022.
- B. Efron and R. J. Tibshirani. *An Introduction to the Bootstrap*. Chapman and Hall/CRC, 1994.
- I. Funke, D. Rivoir, and S. Speidel. Metrics matter in surgical phase recognition. *arXiv:2305.13961*, 2023.
- X. Gao, Y. Jin, Y. Long, Q. Dou, and P.-A. Heng. Trans-SVNet: Accurate phase recognition from surgical videos via hybrid embedding aggregation transformer. In *MICCAI*, 2021.
- Y. Jin, Y. Long, C. Chen, Z. Zhao, Q. Dou, and P.-A. Heng. Temporal memory relation network for workflow recognition from surgical video. *IEEE Trans. Medical Imaging*, 40(7):1911–1923, 2021.
- K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick. Masked autoencoders are scalable vision learners. In *CVPR*, 2022.
- W.-Y. Hong, C.-L. Kao, Y.-H. Kuo, J.-R. Wang, W.-L. Chang, and C.-S. Shih. CholecSeg8k: A semantic segmentation dataset for laparoscopic cholecystectomy based on Cholec80. *arXiv:2012.12453*, 2020.
- G. Ilharco, M. Wortsman, R. Wightman, et al. OpenCLIP, 2021. Zenodo 5143773.
- T. J. M. Jaspers, R. L. P. D. de Jong, Y. Li, C. H. J. Kusters, F. H. A. Bakker, R. C. van Jaarsveld, G. M. Kuiper, R. van Hillegersberg, J. P. Ruurda, W. M. Brinkman, J. P. W. Pluim, P. H. N. de With, M. Breeuwer, Y. Al Khalil, and F. van der Sommen. Scaling up self-supervised learning for improved surgical foundation models. *Medical Image Analysis*, 108:103873, 2026.
- Y. Liu, M. Boels, L. C. Garcia-Peraza-Herrera, T. Vercauteren, P. Dasgupta, A. Granados, and S. Ourselin. SKiT: A fast key information video transformer for online surgical phase recognition. In *ICCV*, 2023.
- Y. Liu, M. Boels, L. C. Garcia-Peraza-Herrera, T. Vercauteren, P. Dasgupta, A. Granados, and S. Ourselin. LoViT: Long video transformer for surgical phase recognition. *Medical Image Analysis*, 99:103366, 2025.
- M. Oquab, T. Darcet, T. Moutakanni, H. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, et al. DINOv2: Learning robust visual features without supervision. *TMLR*, 2024.
- A. Perez, C. Nwoye, R. R. Kermani, O. Mohareri, and M. A. Jamal. SurgLaVi: Large-scale hierarchical dataset for surgical vision-language representation learning. *Medical Image Analysis*, page 103982, 2026.
- A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever. Learning transferable visual models from natural language supervision. In *ICML*, 2021.
- S. Ramesh, V. Srivastav, D. Alapatt, T. Yu, A. Murali, L. Sestini, C. I. Nwoye, et al. Dissecting self-supervised learning methods for surgical computer vision. *Medical Image Analysis*, 88, 2023.
- L. Spencer, S. Wang, and C. Chen. Data-efficient surgical phase segmentation in small-incision cataract surgery: A controlled study of vision foundation models. *arXiv:2604.10514*, 2026.
- S. Schmidgall, J. W. Kim, J. Jopling, and A. Krieger. General surgery vision transformer: A video pre-trained foundation model for general surgery. *arXiv:2403.05949*, 2024.
- A. P. Twinanda, S. Shehata, D. Mutter, J. Marescaux, M. de Mathelin, and N. Padoy. EndoNet: A deep architecture for recognition tasks on laparoscopic videos. *IEEE Trans. Medical Imaging*, 36(1):86–97, 2017.
- Z. Wang, B. Lu, Y. Long, F. Zhong, T.-H. Cheung, Q. Dou, and Y. Liu. AutoLaparo: A new dataset of integrated multi-tasks for image-guided surgical automation in laparoscopic hysterectomy. In *MICCAI*, pages 486–496, 2022.
- Z. Wang, C. Liu, S. Zhang, and Q. Dou. Foundation model for endoscopy video analysis via large-scale self-supervised pre-train. In *MICCAI*, pages 101–111, 2023.
- R. Wightman. PyTorch image models, 2019. <https://github.com/huggingface/pytorch-image-models>.
- K. Yuan, N. Navab, and N. Padoy. Procedure-aware surgical video-language pretraining with hierarchical knowledge augmentation. *NeurIPS*, 37:122952–122983, 2024.

- S. Zhang, Y. Xu, N. Usuyama, H. Xu, J. Bagga, R. Tinn, S. Preston, R. Rao, M. Wei, N. Valluri, et al. A multimodal biomedical foundation model trained from fifteen million image–text pairs. *NEJM AI*, 2(1):AIoa2400640, 2025.
- K. Park, Y. Jeon, S. Lim, S. Park, J. Shin, J. Y. Kim, S. An, J. Rhu, J. Kim, G.-S. Choi, N. Oh, and K.-H. Jung. A generalizable foundation model for intraoperative understanding across surgical procedures. *arXiv:2602.13633*, 2026.